Exploiting SOA for Adaptive and Distributed Manufacturing with Cross Enterprise Shop Floor Commerce

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ABSTRACT
Deploying web services on Shop Floor devices in a manufacturing enterprise extends the Service Oriented Architecture (SOA) to the device level, bringing Shop Floor and Enterprise computing resources closer to each other. In this paper, we further extend the SOA landscape towards cross enterprise scenarios, which benefit adaptive manufacturing to be extremely flexible till the last stages of production. A method using mash up of web service technologies like DPWS, BPEL and eSOA, is employed in this work. Possible methods to implement Web Services on wireless sensor nodes like SunSPOTs are also examined. These methods could be used together as an experimental platform to evaluate a new paradigm of factory automation and enterprise computing, in which shop floor real time data are easily accessible from the business level and can be shared among enterprises. This brings a new way of cross enterprise computing called Shop Floor Commerce.

Categories and Subject Descriptors
C.3 [**Special-Purpose and Application-Based Systems**]: *Real-time and embedded systems, Microprocessor/microcomputer applications, Process control systems.* J.7 [**Computer-aided Engineering**]: *Computer-aided Manufacturing*

General Terms
Design, Standardization, Experimentation, Economics, Management

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SOA for eManufacturing, Hybrid Manufacturing Process, Factory Automation, B2B Manufacturing Integration, Cross-Enterprise Shop Floor Commerce.

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1. INTRODUCTION
As microcontrollers with more complex architectures fit in small packages, computing intensity and large volumes of data are active elements in the industrial automation environment. They serve as ubiquitous resources for fine grained data, which themselves are used extensively by remote locations for analysis and maintenance of large automation installations [2,3].

The near future of the manufacturing industry foresees that software enterprise applications like Enterprise Resource Planning (ERP), which form the top nodes of the IT automation tree, could make use of real time data on the shop floor and report trends, causes and analysis. This shop floor intelligence could be gathered on real time, allowing the businesses to adapt to market demands and forecast shop floor breakdowns in a timely fashion [1,3]. Sharing certain relevant data in a secure way could enable the exchange of real time shop floor data across enterprises, make meaningful decisions based on demands, and have an efficient balance on the fast changing product life cycles.

However, current manufacturing infrastructures cannot support this desire because of the existing integration gap between the shop floor level and the enterprise back-end level. So far, communication between these two levels is often done by human integration, file transfer, or using some proprietary protocols for specific kinds of shop floor devices. The introduction of middleware systems like Manufacturing and Execution Systems (MES) and Distributed Control Systems (DCS) improves the connection and control, but they must also be statically tailored for each group of devices of a certain vendor. Hence, the communication between the shop floor and the back-end in current manufacturing is error-prone, inflexible, and not in a timely manner. Moreover, because of the static configuration of the middleware, the maintenance cost for a variety of different kinds of devices would be very high.

Web Service, as a single communication abstraction in the SOA context, was originally aimed to extend the corporate blocks of homemade commercial transactions across enterprises. They started with web applications on enterprise web servers, and brought more flexibility and interoperability to cross enterprise transactions that require constant changes through innovations to adapt to market demands. Microsoft and a few printer companies extended this idea of web services to embedded devices and proposed standards like Device Profile for Web Services (DPWS) [4]. Several European projects like SiRENA [5], SODA [6], and SOCRADES [7] joined the band wagon and extended the specifications to a low footprint embedded stack that can be directly ported on devices with limited computing resources [8]. As a result, devices can host web services (SOA-ready devices) and proactively participate in the SOA landscape.

Deploying Web Services directly on shop floor devices exploits the SOA paradigm down to the shop floor level, bridging the existing communication gap between the shop floor and the back-end systems in current manufacturing industry. Using uniform communication protocols like SOAP and consistent interface specifications like WSDL enables manufacturing back-end systems to seamlessly interact with devices from different vendors without changing the middleware layer. This brings the shop floor devices closer to novel business processes, which are modeled as an orchestration of web services hosted on hybrid systems within an individual enterprise or even across multiple enterprises to share and effectively use shop floor data.

This paper aims to extend the service oriented landscape vertically down to shop floor level for more flexibility in manufacturing, and horizontally towards cross enterprise scenarios, which benefit adaptive manufacturing to be extremely flexible till the last stages of the production. Section 2 introduces our proposed approach to realize this desire using mash-up of several web service technologies and products. Some advantages are also explored, especially by the new feature of cross-enterprise Shop Floor Commerce. Then, we demonstrate the application of this method for a sample cross-enterprise scenario in Section 3. Lastly, Section 4 summarizes the paper and introduces further related issues that we intend to tackle in future.

2. TOWARDS ADAPTIVE AND DISTRIBUTED MANUFACTURING

This section proposes an approach using current SOA technologies and products to implement an adaptive and distributed manufacturing that can share the shop floor data among enterprises for more flexibility and real world awareness.

2.1 Enterprise SOA for Adaptive Manufacturing

Bringing Web Services on devices, flexibility to the shop floor is heavily exploited by projects like SOCRADES [7] in the European Union Project frameworks. In the SOCRADES project, devices are hosted with the DPWS stack that supports a collection of Web Service Standards such as WS-Addressing, WS-Transfer, WS-Discovery, WS-Eventing, etc. In one of the disseminations, a small footprint C language based DPWS stack is deployed on programmable logic controllers (PLC), which host generic services like starting and stopping a mechanic device like a robotic arm. Another practical implementation of DPWS web services on sensor nodes could be realized by SunSPOTs, which run a small footprint of the java virtual machine Sqwak [9].

The traditional SAP ERP systems use iDOCs to pass data between system landscapes. iDOCs are large XML documents which contain a huge amount of data relevant for business transactions. It is very unlikely that these giant business data would be used by a small microcontroller. Conversely, data coming from a shop floor device would not have mandatory data elements needed for transactions in the ERP system. Although deploying web services on the shop floor makes the software artefacts become more embedded SOA systems, shop floor services are more granular and concentrate only on technical issues. To consume these services and generate more business aspects for the enterprise systems like ERP, Manufacturing and Execution Systems (MES) play a dominant role to act as a middleware gateway. SAP ERP systems also provide services which are harmonized enterprise models based on process components, business objects and global data types (GDTs). They follow well defined common rules of business content.

This enables an ERP transaction to be more atomic, stand alone and a stateless nature. Hence, data can be seamlessly transferred between the shop floor and the ERP systems. A sample MES middleware that makes efficient usage of shop floor data and generate meaningful key performance indicators (KPI) is the SAP Manufacturing Intelligence and Integration (SAP MII) [12]. It gathers data from shop floor web services and encapsulates them with proper syntax that can be clearly understood by SAP ERP products. It can also understand data from the ERP systems more efficiently than devices on the shop floor with low resources. It offers the Shop Floor Manager a comprehensive overview of all shop floor status through a visualization GUI, e.g. which production order is getting executed on a particular machine and the status of components on that machine. The SAP MII is regularly updated with shop floor data, stores them in a database for historical analytical purpose. Intelligent reports can also be generated from manufacturing analytics and performance factors provided by the SAP MII. Unlike most other MES systems, the SAP MII can offer its functionalities over web services. Another advantage of the SAP MII is that it also provides connectors for legacy shop floor devices (non-SOA-devices) to integrate them into the SOA landscape.

SOA context prevails from the shop floor to the enterprise level with different functionalities, forming an enterprise SOA (eSOA) landscape within a manufacturing enterprise. Consisting of cross-layer services which provide raw shop floor data, enterprise level information and manufacturing analytics, such an eSOA could be composed to more large-scale hybrid business processes to achieve higher-level business goals. Currently, Business Process Execution Language (BPEL) [15] seems to be the dominant language to describe executable business processes. The advantage of BPEL is that it is widely deployed in business process management systems and well accepted among business process designers. Although it is mainly built for orchestrating web services, i.e. invoking them, handling results, and using return values as input for other services, it is Turing-complete, meaning it can be used to model any algorithm. The standardized and well-structured characteristics of BPEL encourage us to apply it for cross-layer service composition, constructing hybrid business processes in an eSOA manufacturing landscape, as depicted in Figure 2.1.

The ubiquitous nature of devices and the empowerment of web services on them improve the real world visibility and the awareness of available resources. Distributing business intelligence to shop floor devices and integrating them into enterprise level give eSOA an upper hand on providing flexibility

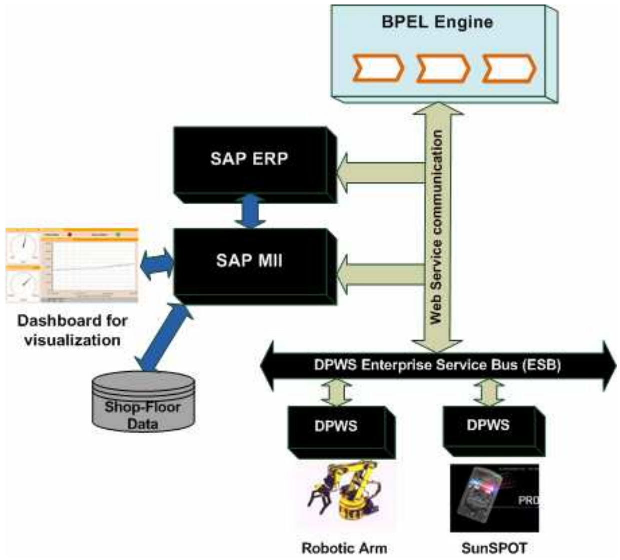

Figure 2.1 Overview of a sample eSOA manufacturing landscape.

to the business house. Exposing the dynamic nature of the shop floor along with metadata enables shopping for shop floor devices. As an example, the Enterprise Service Bus (ESB) in Figure 2.1 allows us to monitor the shop floor device status, to ensure the load balancing among similar devices, and select the suitable devices for certain demands. Hybrid business processes can now be provided with best-fit shop floor devices at the design time, deployment or even dynamically at runtime. It enables production houses to introduce "just in time" characteristics for an adaptive manufacturing, to reduce costs, risks, and manufacturing delay.

For small and medium sized companies that could have customized ERP systems or share enterprise services from a provider like SAP Business By Design server [14], the need for highly expensive middleware components like the MES systems could be eliminated to save costs and at the same time, provide more room for cross company collaborations on services. Upon careful analysis and depending on the business nature, data can be shared across the brick and wall factory to external consumers.

2.2 The Distributed Manufacturing Enterprise

In an effort to improve flexibility in meeting customer demands and leverage technological breakthroughs, several automotive companies have spun off the production of their electronic components [13]. Manufacturing houses struggle to differentiate their products by giving more attention and focuses only on special expertise, and spinning off other parts to multi-located facilities. This leads to decentralized and distributed manufacturing processes that would span over organizations, nations and even continents, forming a Business-To-Business (B2B) model for the manufacturing industry. Such an integrated manufacturing are leaner and more flexible, motivating improved productivity and competitive advantage. The real issue for the manufacturers is to bring the flexibility across their production landscape and respond efficiently on new demands. This grows to be an expectation also on the partners in the value chain.

Because BPEL is a standardized and widely accepted business process language, it might be the most suitable one to construct B2B manufacturing processes across enterprise boundaries. Moreover, exploiting the SOA paradigm on the shop floor level integrates shop floor devices seamlessly and directly into the enterprise level, enabling the visibility of just-in-time shop floor data in a B2B manufacturing process. This brings a new way of cross enterprise computing called Shop Floor Commerce. Manufacturing businesses can now use shared shop floor data and derive the KPIs of all available partner candidates to estimate the profits of collaboration. After analyzing the shop floor data and the KPIs, an ad-hoc decision can be made to select the best business partner among others, and then put him on the manufacturing chain to maximize the business value.

The seamless and direct integration of the shop floor into a B2B process also introduces a timely and flexible manner to handle critical events that may occur in the shop floor at runtime. Business partners can now be informed immediately about the unexpected problems, e.g. delayed delivery or canceled order. Hence, it reduces costs and negative impact on long-term business relation.

3. REFERENCE IMPLEMENTATION

In this section, our approach in the previous Section 2 is applied for a sample B2B production process. The motivating scenario is introduced in subsection 3.1. Then, the detailed implementations are explained in subsections 3.2 and 3.3.

3.1 Motivating Scenario

We considered a scenario where there are two business houses A and B associated with specific business services. Company A for instance, could be an automobile manufacturer located in Europe. Company B might manufacture electronic parts like the climate controller for automobiles, located across the globe in Asia. For an incoming production order, company A initializes a production process and completes the main manufacturing part. After that, the process might cross the organizational boundary to consume the services of a company B to finish the rest parts. However, there might be many B's existing in the market, providing the same services. The first goal is to demonstrate that company A could request its Decision Service to search for a B with specific KPIs like less time to manufacture, high machine availability and no production queues. The selection is done by the Decision Service based on real time data from the shop floors of the B candidates. To achieve this, we propose to deploy SOA till the device level on the shop floor of each company B, calculate KPIs, and expose them over web services so that external companies or partners like A can use them for selecting the best fit one among numerous available firms of type B. For a more accurate comparison, the Decision Service at A can also calculate the KPIs itself, based on the retrieved shop floor data of several B firms. Then, the manufactured goods are transported to the selected company B. The main production process at A triggers a local manufacturing sub-process at B and keeps waiting until this sub-process finishes.

The second goal is to demonstrate the adaptive and flexible manufacturing inside the selected company B, where several shop floor devices can be selected for the manufacturing. The Decision Service at B should identify the best manufacturing device, based on real time shop floor metadata such as availability, manufacturing delay or statistical breakdown rate. After a manufacturing device is selected, its temperature data must be periodically checked whether it is under a certain threshold critical for the manufacturing. The manufacturing at B can start if the average temperature result is under the threshold.

The third and last goal is to simulate the error handling of shop floor events generated at runtime. In case no manufacturing device is available at B or the shop floor temperature at B is too high, the manufacturing process at B cannot be completed. An event is generated and notified to the enterprise level of both B and A. Hence, the main production process at A can be timely aware of the problem at B. It can then move on to other negotiations or look for another company B.

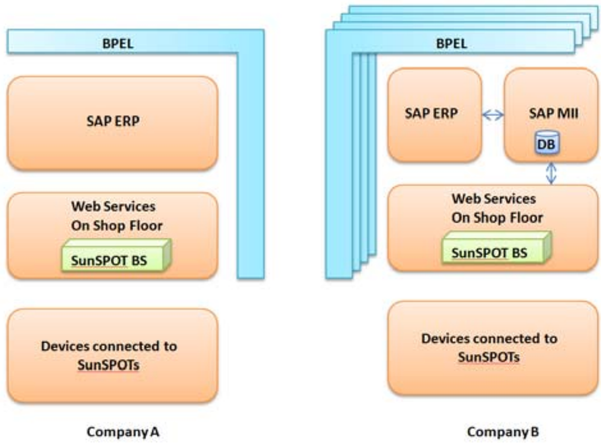

Figure 3.1 Components in our scenario

Figure 3.1 depicts the service components in our scenario. During the whole cross-enterprise production process, status and location of the production order are updated to both two ERP systems of A and B. Shop floor data are regularly updated and visualized in the SAP MII. Intelligent reports and calculated KPIs can be provided by the SAP MII as well. Thus, a business manager can keep track of the running, pending and failed production orders in his house. A Shop Floor Manager can have a comprehensive overview of real time status of shop floor devices and hence, can diagnose the upcoming critical problems.

A Decision Service can be a single back-end service at an enterprise or, as we now assume, embedded in the ERP system. To bring SOA to the shop floor level, we considered a practical implementation of DPWS stack on top of SunSPOT devices.

3.2 SOA at the Shop Floor Level

The used SunSPOTs consist of two Base Stations (BS) and two remote nodes. One BS is terminated to a desktop PC over an USB port labeled A. Meanwhile, a remote node forms a part of the automation hardware on the shop floor. The remote node hosts a set of temperature sensors and accelerometers. It also has 8 GPIO pins and a set of high current drivers to turn on 125mA relays. The relays activate a 12VDC motor. The motor is considered as an electrical device that is most commonly found on the shop floor machines in an automobile manufacturing plant. This setup is considered as a part of the shop floor at a company A. A proximity sensor is terminated to the other remote SunSPOT. Some LED's are also connected to the GPIO pins of the SunSPOT. A similar motor is also connected to the GPIO pins over relays. This SunSPOT is connected to the second base station to another PC and this setup is considered as a company B.

The remote SunSPOTs communicate with their base stations over 2.4GHz wireless channels. The base station at A sends telemetry commands to its remote node to start and stop the motor. The start and stop commands at the base station are encapsulated with web services. Whenever a GPIO pin is triggered on the remote node, it will be forwarded to the base station as a web service event. Other events like malfunctions of the motor or other devices could also be realized in this way. The use of wireless devices like the remote SunSPOTs gives the possibility to place the devices remotely and monitor the performance of these devices under hazardous conditions, which are often seen on the shop floor. The overview of the software components and their associated hardware is seen in Figure 3.2.

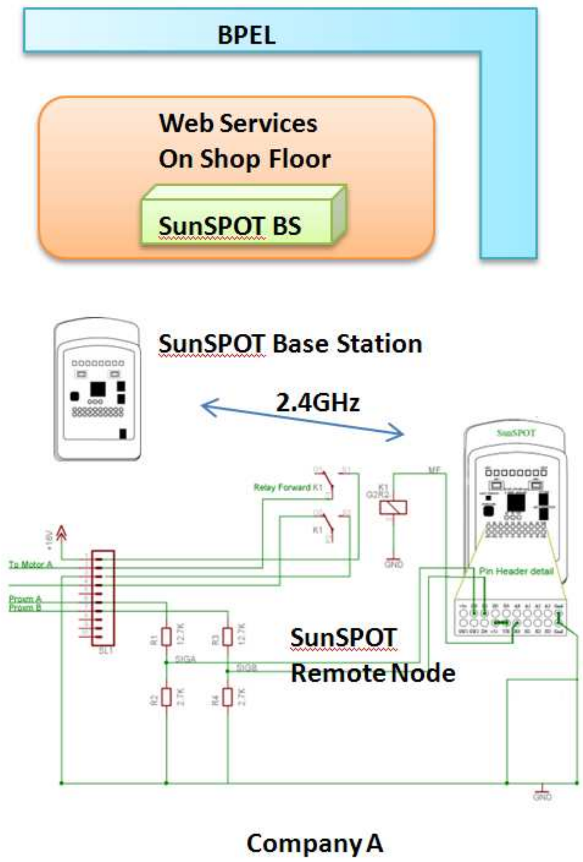

Figure 3.2 Overview of the architecture using SunSPOTS on shop floor

The software used to program the SunSPOTS is written in Java using the NetBeans v6 IDE. The development kit also comes with a SDK, which is currently updated to version 3 called Purple.

Using the SDK, the GPIO ports and the temperature sensors can be accessed using simple APIs like

private ITemperatureInput tempSensor = EDemoBoard.getInstance().getADCTemperature();

The acquired data can then be transmitted using methods like,

Radiogram rdg =
xmit.newDataPacket(PING_REPLY);
rdg.writeInt(linkQuality);

xmit.send(rdg);

Where, *PING_REPLY* is a telemetry command that both the base and remote nodes can understand. It is normally an integer value.

3.3 SOA at the Enterprise Level

Given a completely service oriented manufacturing landscape, we used BPEL to model an experimental hybrid production process across the boundary between the company A and B. In our BPEL process, the SAP MII at B was used for visualization of shop floor data, persisting data in a database for historical purpose, and calculating analytical KPIs for external partners like the company A. Unlike other common MES systems, SAP MII can expose its functionalities as web services to the outside. Figure 3.3 shows a snapshot of the GUI in SAP MII that is used to visualize runtime and historian shop floor data.

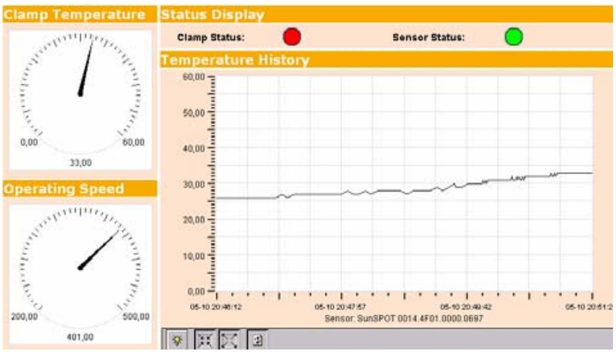

Figure 3.3: The dashboard of SAP MII

Two SAP Supply Chain Management Production Planners (SCM PP) were used as ERP systems involved in the BPEL process, one at the company A and one at the company B, in order to keep track of the status and location of all production orders. Since traditional SAP systems can only be accessed via iDOCs or Remote Function Calls (RFC), we implemented two proxy web services for these ERP systems that wrap the RFC procedures by web service operations. In this way, status and location of production orders can be updated to ERP systems directly from the BPEL process over web service calls.

We simplified the runtime calculations of KPIs by introducing a virtual cost assigned to each candidate B. The decision service in A simply chooses the cheapest B. Similarly, selecting a manufacturing device inside B was also implemented with virtual costs.

Business rules were modeled in the BPEL process, e.g. temperature check and event handling. To intentionally trigger a shop floor event, we used a dryer to heat the remote SunSPOT device in a period of time. The high temperature values were periodically sent back to the SunSPOT base station. Then, they were wrapped in web service events generated by DPWS, and the events were notified to the BPEL process. In the BPEL process, an event handler was modeled to receive the temperature events, calculate the average temperature and act appropriately.

4. SUMMARY AND FUTURE WORK

The recent trend in the manufacturing industry is that shop floor ubiquitous devices are becoming more powerful regarding the computing and communication resources, up to the point where they can directly offer their functionalities as web services. This paper addresses the feasibility of integrating shop floor devices directly and seamlessly into enterprise level, providing more flexibility and real world awareness in a manufacturing house. Further, this paper extends the SOA landscape to the B2B context, where enterprises can share and exchange shop floor data. This enables more flexible and profitable collaborations, forming a new innovative strategy called Shop Floor Commerce. The proposed experimental method using current SOA technologies and products was successfully applied for a B2B scenario in the automobile domain.

This work belongs to a larger project, aiming at the design and implementation of a comprehensive Smart Item Infrastructure for future manufacturing. Integration of SOA-ready devices into the enterprise domain must face the limitations of the dynamic nature of ubiquitous devices, e.g. mobility and unreliability. More functionalities such as dynamic service mapping and deployment, device and service monitoring, discovery, etc should be supported by the integration. These issues can be tackled with the more capable middleware integration infrastructure introduced in [1]. The successful design and implementation of the prototype approach in this paper encourage us to move towards the complete implementation of that middleware. Regarding technical issues, our partners in the project are trying to deploy the DPWS stack on several resource constrained platforms, e.g. directly on a SunSPOT base station which runs a Sqwak virtual machine.

In a shop floor, there might be hundreds of automation hardware to be started or stopped at a particular time. Exposing them all as Web Services would add more chaos to the shop floor. Hence, the Web Services need to be added with more context and semantic information, in order to easily distinguish different events and places of the shop floor. In B2B collaboration, semantic annotations for exchanged shop floor data would ensure the interoperability among multi-located partners. At this point, we are slowly moving to the world of semantic web service description for manufacturing domain.

In [10] Spiess et al. proposed the idea of partitioning a BPEL process into sub-processes and distribute them down to the shop floor level, to increase the locality and reliability of the manufacturing processes. We also intend to integrate this innovative idea into our next experimental prototype. However, the challenge is how to find a suitable light-weight process engine that can run on a resource-constrained device and execute a distributed sub-process.

The scenario described in this paper is just a simple prototype. Manufacturing processes in the reality are more complex and could involve hundreds of heterogeneous SOA-ready shop floor devices. In order to examine the runtime behaviors and assess the operational performance of those complex processes, the simulation approach in [11] could be involved.

5. ACKNOWLEDGEMENT

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